

Gastroparesis in the Chronic Kidney Disease Patient: Clinical Management and Implications for Practice



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GASTROPARESIS IS A chronic condition characterized by a delayed gastric emptying in the absence of mechanical obstruction.¹ Although most instances of gastroparesis are seen in individuals with diabetes mellitus and idiopathic subtypes, there are some instances in which chronic kidney disease (CKD) is a comorbidity with this condition. In CKD there is a clear underreporting of digestive pathologies due to the lack of knowledge of the mechanisms that lead to a wide range of symptoms.² Some symptoms that can be relatively similar between gastroparesis and CKD include nausea, vomiting, and early satiety. Just as digestive pathologies in kidney disease can be unclear, determining the presence of gastroparesis can be just as challenging. Gastroparesis is traditionally diagnosed using a solid-phase gastric emptying scan also known as gastric emptying scintigraphy.¹ This noninvasive, quantitative method involves eating a radiolabeled solid meal. Gastric counts, which correlate with the amount of meal remaining in the stomach, are measured scintigraphically at particular time points after the meal is consumed.¹

When it comes to nutrition management of gastroparesis and CKD, there are several recommendations that can support symptoms with both disease states. The first recommendation to consider is encouraging more small and frequent meals. Consuming smaller portions with more frequent meals lessens the amount of food in the digestive system and can support the individual with slow gastric emptying. With CKD, recommendations for protein at these meals should reflect the diet prescription and stage of CKD to prevent kidney burden. Another area of nutrition management is intake of specific micronutrients. With CKD and gastroparesis, many individuals are at risk for

nutrient deficiencies due to inadequate oral intake. The most common include iron, vitamin B₁₂, vitamin D, and calcium.³ For kidney disease, supplements are usually given to patients and may include these specific vitamins and minerals.³ Fat and fiber consumption are two additional areas that need to be considered with gastroparesis. Fat intake slows down stomach emptying, but many individuals can better tolerate liquid fat sources.³ Fiber can also lead to slower digestion which may contribute to gastroparesis symptoms.

More specific recommendations to the kidney diet also need to be considered with this patient population to optimize kidney function. Sodium, typically limited to 2,300 mg/day, should be restricted to help decrease blood pressure and risk for fluid retention. Additionally, two other nutrients to monitor with kidney disease include phosphorus and potassium. With kidney disease, the kidneys' lack of filtration limits removal of phosphorus and increased levels can weaken bones, making it important to limit high amounts of this nutrient.³ As for potassium, elevated levels can cause a rapid heartbeat, increasing the risk of a heart attack. Overall, smaller, more frequent meals and watching portion sizes to ensure recommended amounts of macronutrients and micronutrients are consumed, is important for the patient with both gastroparesis and CKD.

Due to pre-existing pathophysiological complications and potential nutrient deficiencies in patients with gastroparesis and kidney disease, proper nutritional management of symptoms is imperative to optimize nutrient stores and prevent further health decline. Those with CKD often suffer from fluid and electrolyte disturbances which can be further exacerbated when paired with the fluid and electrolyte imbalances that can result from gastroparesis; a severe complication potentiating harmful and possibly fatal outcomes.⁴ Thus, targeted clinical management strategies including use of gastric motility agents and nutrition support may be needed when treating the CKD patient with gastroparesis who is unable to tolerate an oral diet.⁵⁻⁷

In the situations for which standard dietary interventions alone are not able to provide sufficient symptomatic relief, pharmacological therapy is often the next level of treatment

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in an effort to increase tolerance to oral intake.^{5,6} Provision of prokinetic agents to promote peristalsis in conjunction with antiemetics has proven an effective treatment method through facilitation of muscle contraction and alleviation of nausea and vomiting.^{5,6} It should be noted that liquid medications versus tablets/capsules may provide more effective relief due to the quicker rate of passage through the stomach.^{1,6,7} It is also important to consider possible medication interactions and contraindications when administering these agents.⁵⁻⁷

Some patients may require a more advanced level of care. Weight should be carefully monitored, along with evaluation of malnutrition with validated, noninvasive tools such as the Nutrition Focused Physical Exam and the Subjective Global Assessment for assessment based on standard malnutrition criteria.⁶⁻⁹ If such criteria are met and the patient's goal weight cannot be sustained via oral nutrition and pharmacological therapy, initiation of nutrition support may be warranted. Enteral nutrition (EN) or tube feeding is the preferred method of nutrition support and parenteral nutrition should be considered only in the case of extensive intestinal dysmotility after EN is deemed unsuccessful.^{6,7} Jejunal feedings via a percutaneous endoscopic transgastric jejunostomy have demonstrated success in terms of EN provision in kidney disease patients with gastroparesis due to the ability to vent the stomach while allowing suitable delivery of nutrients and medications.^{6,7} In general, special formula is not often necessary as standard EN formula is typically well tolerated in these patients. However, consideration should be given to patients with more significant dysmotility for which fiber containing formula may aggravate gastrointestinal symptoms despite being provided into the small intestine.^{6,7} Gradual increase of feeds to goal from an initial hypocaloric state may be useful for prevention of refeeding syndrome mediated by severe shifts in fluid and electrolytes.¹⁰

Patients struggling with both CKD and gastroparesis are at a heightened risk for various health complications and overall nutritional degradation due to the implications of their pathophysiological processes. In an effort to achieve optimal nutrition and preserve kidney integrity, assessment of the patient's individual needs must be carried out using good clinical judgment coupled with a multidisciplinary approach to care. Whether treated with standard therapeutic interventions or more advanced clinical management such as nutrition support, management of symptoms that inhibits the patient's health is paramount. The following handout provides a quick reference for the patient, including helpful tips for managing gastroparesis symptoms in the context of CKD.

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MANAGING YOUR GASTROPARESIS WITH KIDNEY DISEASE

Managing your diet can be a challenge with kidney disease, and gastroparesis can make it even harder to meet your nutritional needs. Follow these guidelines for some helpful tips for managing your gastroparesis to keep you and your kidneys healthy.



WHAT IS GASTROPARESIS?

Gastroparesis is a condition that affects the passage of food through your stomach. If you have gastroparesis, food digestion may be slowed or delayed causing uncomfortable symptoms.

SIGNS & SYMPTOMS

- Feeling full after only a few bites
- Bloating
- Nausea & vomiting
- Decreased appetite
- Abdominal pain
- Weight Loss
- Acid reflux



TIPS FOR MANAGING YOUR GASTROPARESIS

Rethink Your Portions

Eat smaller portions of food more frequently throughout the day. Eating large portions can slow the emptying process and make symptoms of gastroparesis worse.

Limit The Fat

High-fat or greasy foods may slow down the stomach emptying process.

- ▶ High-fat foods pack a lot of calories. If experiencing weight loss and additional calories are needed, try liquid rather than solid fat sources.

Liquids Vs. Solids

Liquids pass through your stomach faster than solids and may be easier to tolerate. If solid food causes discomfort, try getting nutrients from liquids like fruit and vegetable smoothies or nutrition supplement shakes.

Sit Up & Stay Up

Slouching or laying down while you eat can slow down digestion and aggravate gastroparesis symptoms. Instead, sit up while you eat and avoid slouching or laying down for 1-2 hours after your meal.

Get Active

Exercise can help speed up the digestion process and make it easier for your food to pass through your stomach. If you experience discomfort after a meal, try taking a walk or going on a bike ride.



IF YOU FOLLOW A RENAL DIET, IT IS IMPORTANT TO MAKE SURE YOU CONTINUE TO FOLLOW THE RENAL DIET GUIDELINES. TALK TO YOUR DOCTOR OR SET UP AN APPOINTMENT WITH A REGISTERED DIETITIAN TO DISCUSS YOUR SPECIFIC NEEDS AND TO MAKE SURE YOU ARE GETTING ENOUGH NUTRIENTS IN YOUR DIET!